

# AKARSH SHAURYA

**COMPUTER SCIENCE & ENGINEERING UNDERGRADUATE**

Backend engineering | Edge AI | Thoughtful product experiences

akarshshaurya-dev.vercel.app
github.com/WDuck45
VIT Vellore, India

## EDUCATION

**Vellore Institute of Technology**
**B.Tech, Computer Science & Engineering**
Vellore, India
**Expected graduation: July 2030**

**Coursework**
Harvard CS50 (programming foundations)
Multivariable Calculus & Differential Equations, with GNU Octave lab

## TOOLKIT

**Core Engineering**

Python | FastAPI | MySQL | REST APIs | C / C++ | Git | GitHub | Terminal | macOS (APFS)

**AI & Signal Processing**

Feature Extraction | Higuchi Fractal Dim. | Hjorth Complexity | Random Forest | Grid Search CV | SHAP

**Scientific Computing**

GNU Octave | MATLAB | Symbolic Calculus | Numerical Analysis | Partial Derivatives

**Creative & Craft**

DaVinci Resolve | Keyframe Animation | Visual Rhythm

## HOW I THINK

**Inspect, reason, and refine.**
I prefer tracing problems down to terminal logs and signal inputs over surface-level assumptions. Systems should be built for reliability under the hood, with equal care given to visual pacing and user intuition.

## PROFILE

First-year CSE student at VIT Vellore engineering scalable backend architectures and exploring edge-AI applications for real-world biosignals. Currently leads signal processing and ML modeling for a PPG-based sleep monitor, combining core backend engineering with a strong eye for visual rhythm and user experience.

## PROJECTS

### Edge-AI Wearable for Sleep & Apnea Risk Assessment

Multidisciplinary Project (MDP)

*Lead AI & Signal Pipeline Developer*

- Building a real-time signal pipeline that extracts and fuses 20+ statistemporal and nonlinear features (Higuchi Fractal Dimension, Hjorth Complexity) from raw 30-second optical PPG pulse waveforms.
- Training explainable Random Forest models tuned with Grid Search CV, targeting ~82% + accuracy for non-invasive sleep staging and pattern flags.
- Integrating SHAP (SHapley Additive exPlanations) for clinical transparency, mapping every prediction back to underlying physiological feature attributions.

### SmartSplit AI

Hackulus Team Project

*Backend Developer*

- Engineered the FastAPI backend and relational MySQL database schemas for group expense management, integrating microservices with a React frontend.
- Implemented automated UPI deep-link payment generation alongside JWT authentication to secure user sessions and API endpoints.

`github.com/WDuck45/SmartSplit-AI`

### LINKVIT: VIT Networking & Opportunities Platform

Collaborative Team Project

*System Architect, Backend & Database Design*

- **Designed** the system architecture and database schemas for domain-based peer matching and opportunity aggregation within the VIT ecosystem.
- **Modeled** storage for member profiles, education details and connections in a LinkedIn-style networking app.

## ACHIEVEMENTS & COMMUNITY

- **National Olympiads (2025):** qualified and certified in the Indian Olympiad Qualifier in Mathematics (IOQM) and the National Standard Examination in Chemistry (NSEC).
- **Hackathons & events:** Hackulus (built SmartSplit AI with a team); active participant in Clueminati 4.0 (CodeChef-VIT).
- **Student communities:** member of VIT's RoboVITics and AI/ML clubs.